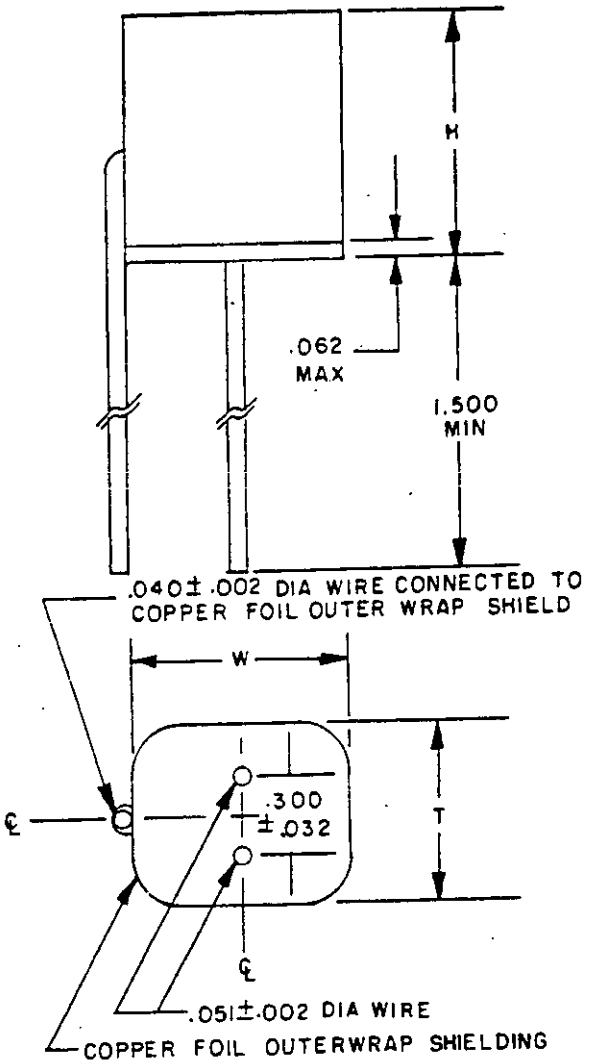
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, METALLIZED PLASTIC DIELECTRIC, DC,
IN NONMETAL CASES, ESTABLISHED RELIABILITY
STYLE CFR29

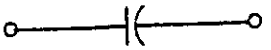
This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-55514.



Style CFR29

Inches	mm
.002	0.05
.040	1.02
.046	1.17
.051	1.30
.062	1.57
.093	2.36
.300	7.62
.437	11.10
.468	11.89
.500	12.70
.562	14.27
.593	15.06
.718	18.24
.750	19.05
.781	19.84
.862	21.89
.875	22.23
.906	23.01
.937	23.80
.968	24.59
1.000	25.40
1.031	26.19
1.093	27.76
1.250	31.76
1.281	32.54
1.312	33.32
1.375	34.93
1.437	36.50
1.500	38.10
1.687	42.85
1.812	46.02



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.

FIGURE 1. Configuration and dimensions.

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TABLE I. Capacitor characteristics and dimensions.

Type designation 1/	Capacitance	Capacitance tolerance	Rated voltage at 85°C	ESR milliohms at 100 kHz	Ripple current max at 25°C	Resonant frequency kHz	Dimensions T 2/	W 2/	H +.093 -.062
CFR29RRA106--	10.0 µF	F G J K	50 V	11	12.6	318	.562	.862	.750
CFR29RRA206--	20.0 µF	" " " "	50 V	10	14.8	214	.718	.968	.875
CFR29RRA506--	50.0 µF	" " " "	50 V	6	23.5	130	1.000	1.375	1.000
CFR29RRG305--	3.0 µF	F G J K	75 V	15	10.3	672	.468	.718	.750
CFR29RRG505--	5.0 µF	" " " "	75 V	13	11.6	427	.500	.750	.875
CFR29RRG106--	10.0 µF	" " " "	75 V	12	13.5	301	.718	1.031	.875
CFR29RRG206--	20.0 µF	" " " "	75 V	11	16.5	205	.937	1.312	1.000
CFR29RRG506--	50.0 µF	" " " "	75 V	6	24.2	120	1.312	1.687	1.250
CFR29RRB305--	3.0 µF	F G J K	100 V	15	10.6	684	.437	.750	.875
CFR29RRB505--	5.0 µF	" " " "	100 V	13	12.4	427	.593	.906	.875
CFR29RRB106--	10.0 µF	" " " "	100 V	12	14.6	290	.781	1.093	1.000
CFR29RRB206--	20.0 µF	" " " "	100 V	11	17.1	192	.906	1.281	1.250
CFR29RRB506--	50.0 µF	" " " "	100 V	6	23.6	115	1.437	1.812	1.375

1/ The complete type designation will include additional symbols to indicate capacitance tolerance and failure rate level (M, P, R, and S).
2/ Tolerances ±.046 thickness and width of .437 through .500 inches; ±.062 thickness and width of .562 and greater.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case: Nonhermetic.

Dielectric: Metallized polycarbonate.

Characteristic: R.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +85°C, when properly voltage derated (see figure 2), characteristic "R" capacitors will operate to +125°C.

Voltage rating: See table I.

Marking: In accordance with MIL-C-55514.

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Qualification submittal:

Type designation	Sample size
CFR29RRA106FM	61
CFR29RRA506FM	61
CFR29RRG305FM	61
CFR29RRG506FM	61
CFR29RRB305FM	61
CFR29RRB506FM	61

Substitutability for failure rate levels shall be as follows:

Failure rate level	Will replace failure rate level
S	R, P, M
R	P, M
P	M
M	

Substitutability for capacitance tolerance shall be as follows:

Capacitance tolerance	Will replace capacitance tolerance
F	G, J, K
G	J, K
J	K
K	

Substitutability for voltage shall be as follows:

Volts dc	Will replace voltages
100	75, 50
75	50

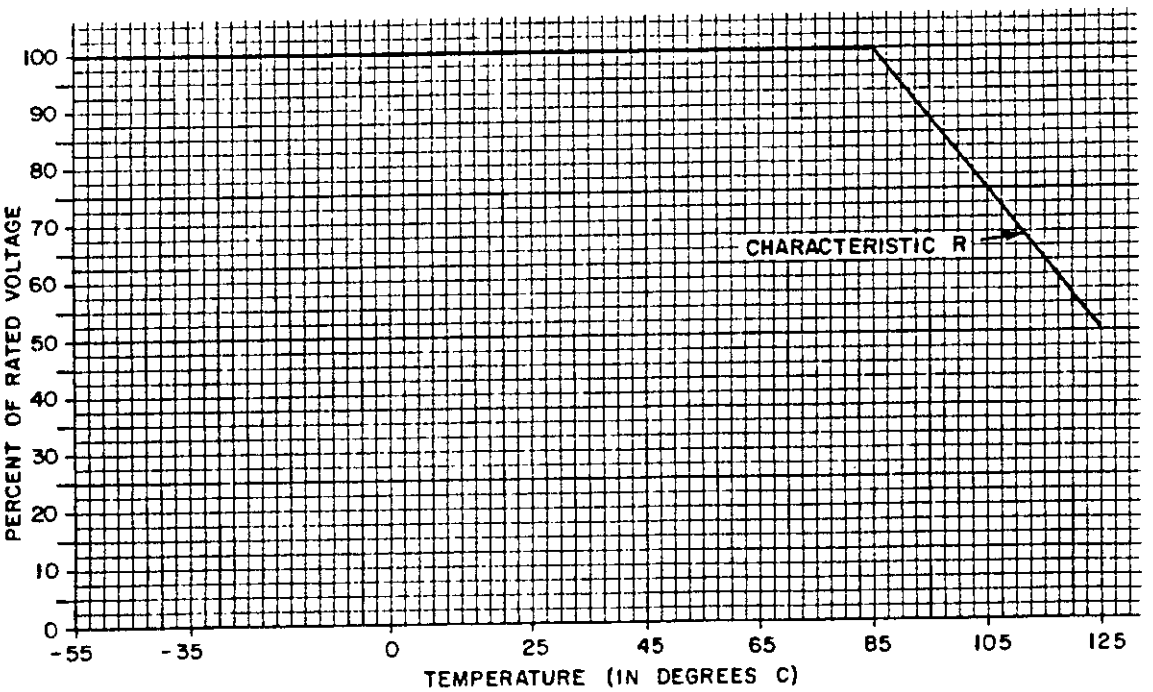


FIGURE 2. Voltage derating with temperature.

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